



Points, Vectors, Lines, Spheres and Matrices

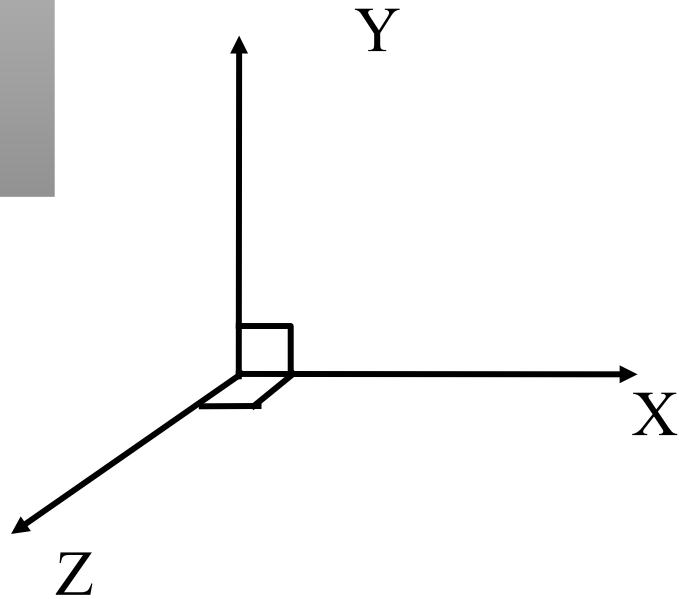
Overview

- *Basic Math*
 - *Points*
 - *Vectors*
 - *Lines*
 - *Spheres*
 - *Matrices*
- *Simple Camera*
 - *Scenes with spheres*
 - *COP on +z*
- *Local illumination*
 - *Ambient*
 - *Diffuse*
 - *Specular*

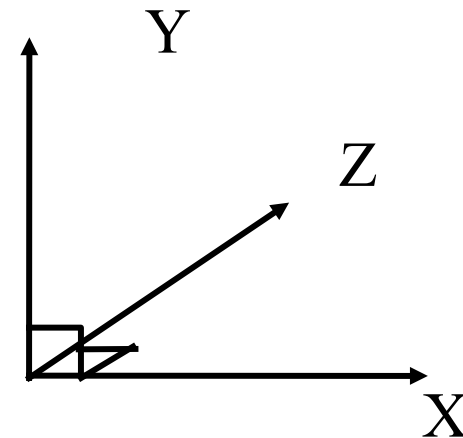
Basic Maths

- *In computer graphics we need mathematics both for describing our scenes and also for performing operations on them, such as projection and various transformations.*
- *Coordinate systems (right- and left-handed), serve as a reference point.*
- *3 axes labelled x , y , z at right angles.*

Co-ordinate Systems



Right-Handed System
(Z comes out of the screen)



Left-Handed System
(Z goes in to the screen)

Points, $P(x, y, z)$

- *Gives us a position in relation to the origin of our coordinate system*

Vectors, $V(x, y, z)$

- *Represent a direction (and magnitude) in 3D space*

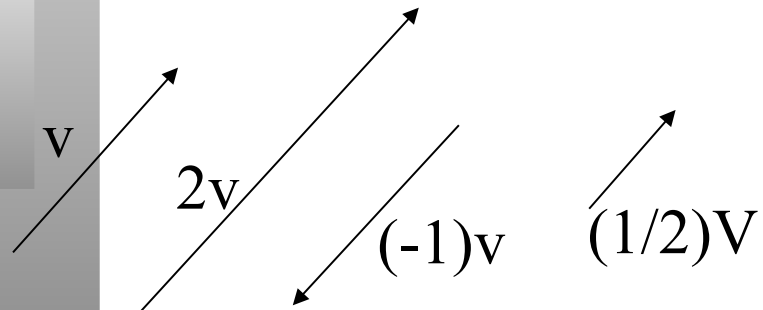
- *Points $\neq$ Vectors*

Vector + Vector = Vector

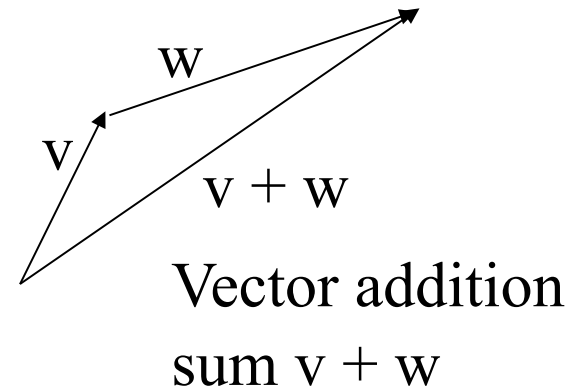
Point – Point = Vector

Point + Vector = Point

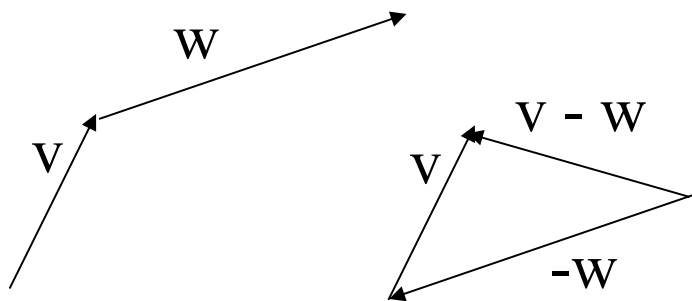
Vectors, $V(x, y, z)$



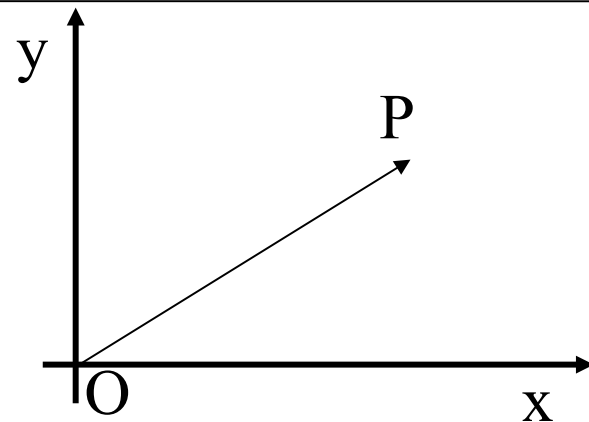
Scalar multiplication of
vectors (they remain parallel)



Vector addition
sum $v + w$



Vector difference
 $v - w = v + (-w)$



Vector $\overrightarrow{OP}$

Vectors $\underline{V}$

- *Length (modulus) of a vector $\underline{V}$ (x, y, z)*

$$|\underline{V}| =$$

$$\sqrt{x^2 + y^2 + z^2}$$

- *A unit vector: a vector can be normalised such that it retains its direction, but is scaled to have unit length:*

$$\hat{V} = \frac{\text{vector } V}{\text{modulus of } V} = \frac{\underline{V}}{|\underline{V}|}$$

Dot Product

$$u \cdot v = x_u \cdot x_v + y_u \cdot y_v + z_u \cdot z_v$$

$$u \cdot v = |u| |v| \cos \theta$$

$$\cos \theta = \frac{u \cdot v}{|u| \times |v|}$$

- *This is purely a scalar number not a vector.*
- *What happens when the vectors are unit*
- *What does it mean if dot product == 0 or == 1?*

Cross Product

- *The result is not a scalar but a vector which is normal to the plane of the other 2*
- *Can be computed using the determinant of:*

$$\begin{vmatrix} i & j & k \\ x_v & y_v & z_v \\ x_u & y_u & z_u \end{vmatrix} \quad \begin{vmatrix} i & j & k \\ x_v & y_v & z_v \\ x_u & y_u & z_u \end{vmatrix} \quad \begin{vmatrix} i & j & k \\ x_v & y_v & z_v \\ x_u & y_u & z_u \end{vmatrix}$$

$$v \times u = \mathbf{i}(y_v z_u - z_v y_u), -\mathbf{j}(x_v z_u - z_v x_u), \mathbf{k}(x_v y_u - y_v x_u)$$

- *Size is $v \times u = |v||u|\sin\theta$*
- *Cross product of vector with itself is null*

Parametric equation of a line (ray)

Given two points $P_0 = (x_0, y_0, z_0)$ and $P_1 = (x_1, y_1, z_1)$ the line passing through them can be expressed as:

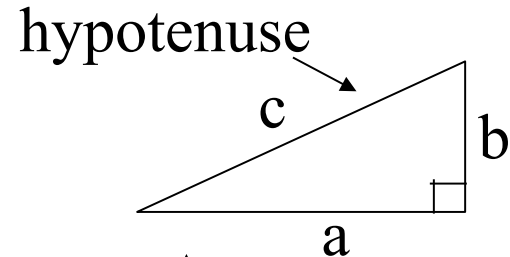
$$P(t) = P_0 + t(P_1 - P_0) = \begin{cases} x(t) = x_0 + t(x_1 - x_0) \\ y(t) = y_0 + t(y_1 - y_0) \\ z(t) = z_0 + t(z_1 - z_0) \end{cases}$$

With $-\infty < t < \infty$

Equation of a sphere

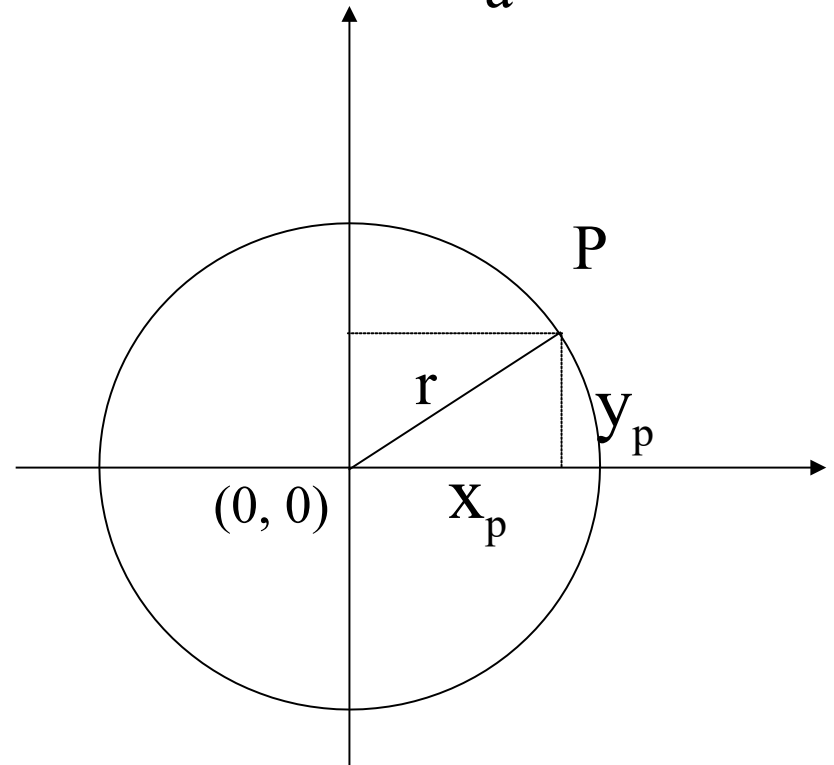
- *Pythagoras Theorem:*

$$a^2 + b^2 = c^2$$



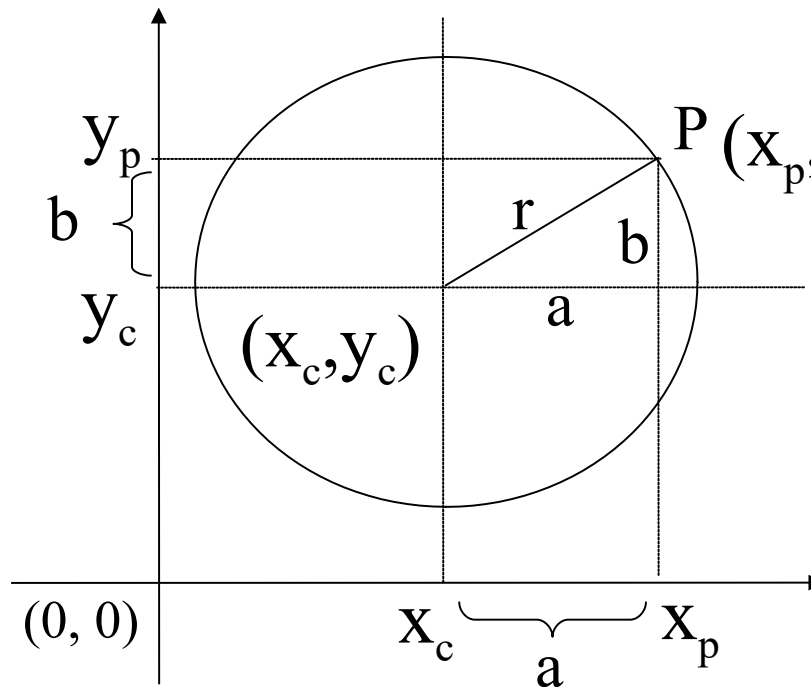
- *Given a circle through the origin with radius r , then for any point P on it we have:*

$$x^2 + y^2 = r^2$$



Equation of a sphere

* If the circle is not centred on the origin:



We still have

$$a^2 + b^2 = r^2$$

but

$$a = x_p - x_c$$

$$b = y_p - y_c$$

So for the general case $(x - x_c)^2 + (y - y_c)^2 = r^2$

Equation of a sphere

- * Pythagoras theorem generalises to 3D giving

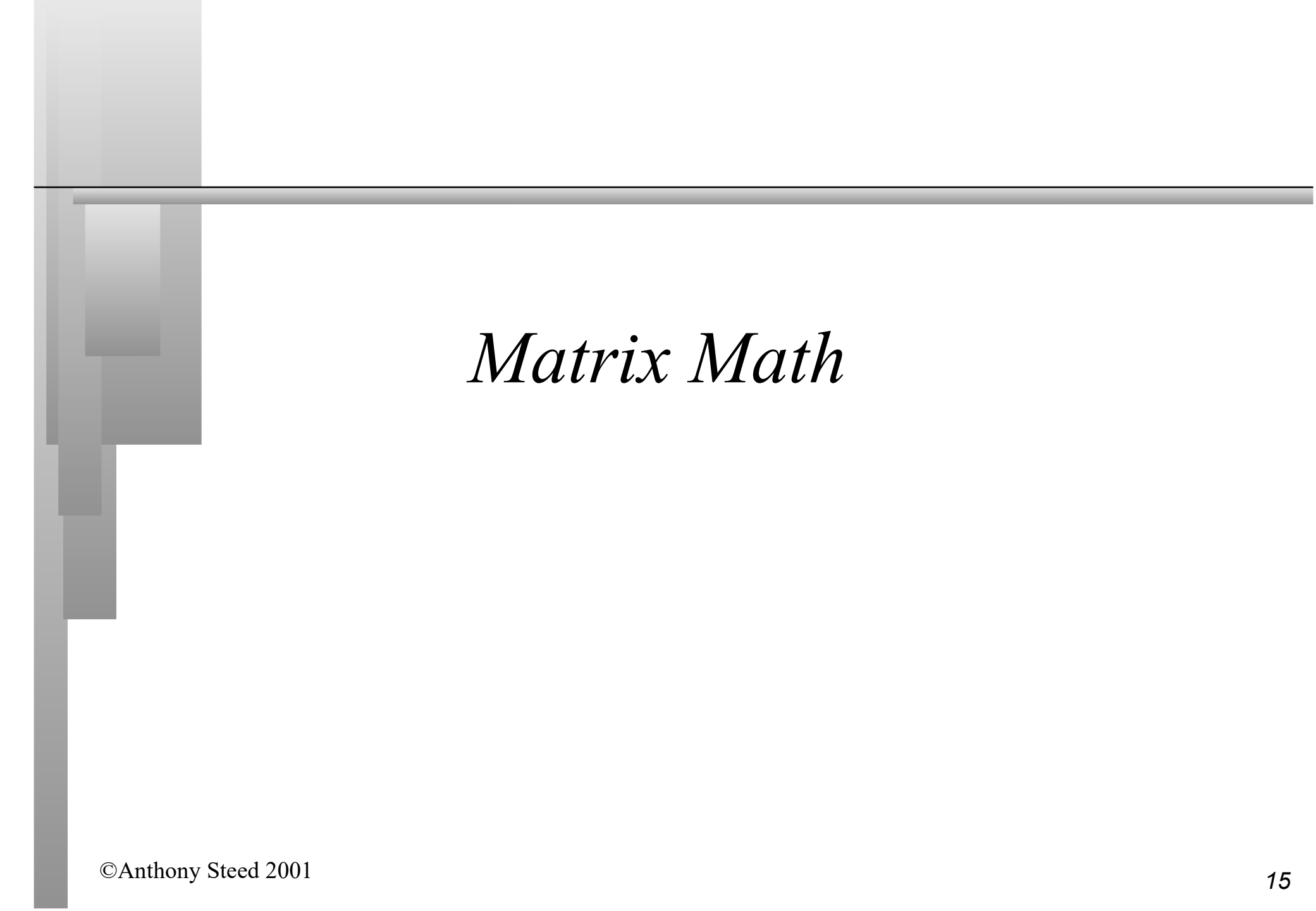
$$a^2 + b^2 + c^2 = d^2 \quad \text{Based on that we can easily}$$

prove that the general equation of a sphere is:

$$(x - x_c)^2 + (y - y_c)^2 + (z - z_c)^2 = r^2$$

and at origin:

$$x^2 + y^2 + z^2 = r^2$$

A decorative graphic consisting of a thick gray vertical bar on the left side of the slide and a thin gray horizontal line extending across the top.

Matrix Math

Vectors and Matrices

- *Matrix is an array of numbers with dimensions M (rows) by N (columns)*
 - *3 by 6 matrix*
 - *element 2,3 is (3)*
- $$\begin{pmatrix} 3 & 0 & 0 & -2 & 1 & -2 \\ 1 & 1 & 3 & 4 & 1 & -1 \\ -5 & 2 & 0 & 0 & 0 & 1 \end{pmatrix}$$
- *Vector can be considered a $1 \times M$ matrix*
$$v = (x \ y \ z)$$

Types of Matrix

- *Identity matrices - I*

$$\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \quad \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

- *Diagonal*

$$\begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 2 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & -4 \end{pmatrix}$$

- *Symmetric*

$$\begin{pmatrix} a & b & c \\ b & d & e \\ c & e & f \end{pmatrix}$$

- *Diagonal matrices are (of course) symmetric*
- *Identity matrices are (of course) diagonal*

Operation on Matrices

- *Addition*

- *Done elementwise*

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix} + \begin{pmatrix} p & q \\ r & s \end{pmatrix} = \begin{pmatrix} a+p & b+q \\ c+r & d+s \end{pmatrix}$$

- *Transpose*

- “Flip” (*M by N becomes N by M*)

$$\begin{pmatrix} 1 & 4 & 9 \\ 5 & 2 & 8 \\ 6 & 7 & 3 \end{pmatrix}^T = \begin{pmatrix} 1 & 5 & 6 \\ 4 & 2 & 7 \\ 9 & 8 & 3 \end{pmatrix}$$

Operations on Matrices

- *Multiplication*
 - *Only possible to multiply of dimensions*
 - x_1 by y_1 and x_2 by y_2 iff $y_1 = x_2$
 - resulting matrix is x_1 by y_2
 - e.g. Matrix A is 2 by 3 and Matrix by 3 by 4
 - resulting matrix is 2 by 4
 - Just because $A \times B$ is possible doesn't mean $B \times A$ is possible!

Matrix Multiplication Order

- A is n by k , B is k by m
- $C = A \times B$ defined by

$$c_{ij} = \sum_{l=1}^k a_{il}b_{lj}$$

- $B \times A$ not necessarily equal to $A \times B$

$$\begin{pmatrix} * & * & * & * & * \\ & & & & \end{pmatrix} \begin{pmatrix} * \\ * \\ * \\ * \\ * \end{pmatrix} = \begin{pmatrix} * \\ \\ \\ \end{pmatrix}$$

$$\begin{pmatrix} * & * & * & * & * \\ & & & & \end{pmatrix} \begin{pmatrix} \cdot & * \\ \cdot & * \\ \cdot & * \\ \cdot & * \\ \cdot & * \end{pmatrix} = \begin{pmatrix} \cdot & * \\ \\ \\ \end{pmatrix}$$

$$\begin{pmatrix} \cdot & \cdot & \cdot & \cdot & \cdot \\ * & * & * & * & * \end{pmatrix} \begin{pmatrix} \cdot & * \\ \cdot & * \\ \cdot & * \\ \cdot & * \\ \cdot & * \end{pmatrix} = \begin{pmatrix} \cdot & \cdot \\ \cdot & * \\ \\ \end{pmatrix}$$

Example Multiplications

$$\begin{pmatrix} 2 & 3 & 1 \\ -1 & 0 & 1 \end{pmatrix} \begin{pmatrix} 2 & 1 \\ 1 & -1 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} _ & _ \\ _ & _ \end{pmatrix}$$

$$\begin{pmatrix} 2 & -2 & 3 \\ -3 & 1 & 0 \\ 1 & -1 & -1 \end{pmatrix} \begin{pmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix} = \begin{pmatrix} _ & _ & _ \\ _ & _ & _ \\ _ & _ & _ \end{pmatrix}$$

Inverse

- *If $A \times B = I$ and $B \times A = I$ then
 $A = B^{-1}$ and $B = A^{-1}$*

3D Transforms

- *In 3-space vectors are transformed by 3 by 3 matrices*

$$(x \quad y \quad z) \begin{pmatrix} a & b & c \\ d & e & f \\ g & h & i \end{pmatrix} = (xa + yd + zg \quad xb + ye + zh \quad xc + yf + zi)$$

Scale

- *Scale uses a diagonal matrix*

$$(x \ y \ z) \begin{pmatrix} a & 0 & 0 \\ 0 & b & 0 \\ 0 & 0 & c \end{pmatrix} = (xa \ yb \ zc)$$

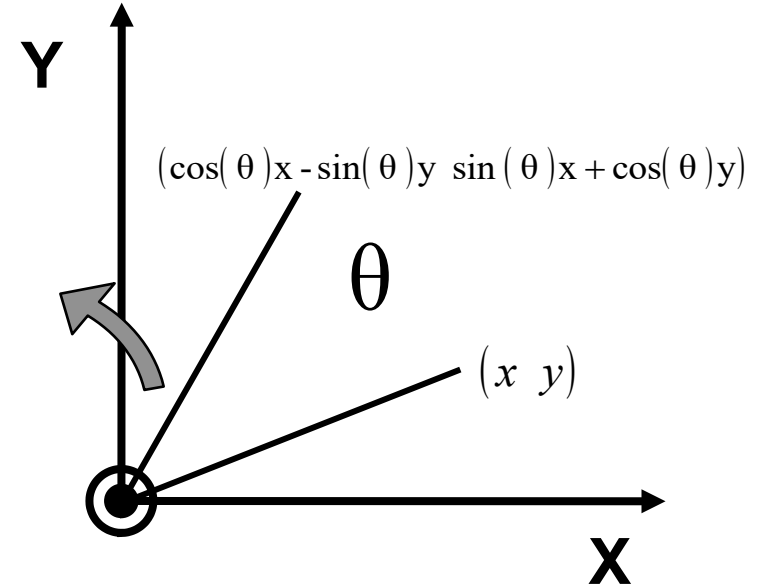
- *Scale by 2 along x and -2 along z*

$$(3 \ 4 \ 5) \begin{pmatrix} 2 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -2 \end{pmatrix} = (6 \ 4 \ -10)$$

Rotation

- *Rotation about z axis*

$$\begin{pmatrix} \cos(\theta) & \sin(\theta) & 0 \\ -\sin(\theta) & \cos(\theta) & 0 \\ 0 & 0 & 1 \end{pmatrix}$$



- *Note **z values** remain the same while **x and y** change*

Rotation X, Y and Scale

- *About X*

$$\begin{pmatrix} 1 & 0 & 0 \\ 0 & \cos(\theta) & \sin(\theta) \\ 0 & -\sin(\theta) & \cos(\theta) \end{pmatrix}$$

- *About Y*

$$\begin{pmatrix} \cos(\theta) & 0 & -\sin(\theta) \\ 0 & 1 & 0 \\ \sin(\theta) & 0 & \cos(\theta) \end{pmatrix}$$

- *Scale (should look familiar)*

$$\begin{pmatrix} a & 0 & 0 \\ 0 & b & 0 \\ 0 & 0 & c \end{pmatrix}$$

Homogenous Points

- *Add 1D, but constrain that to be equal to 1
(x,y,z,1)*
- *Homogeneity means that any point in 3-space can be represented by an infinite variety of homogenous 4D points*
 - $(2\ 3\ 4\ 1) = (4\ 6\ 8\ 2) = (3\ 4.5\ 6\ 1.5)$
- *Why?*
 - *4D allows us to include 3D translation in matrix form*

Homogenous Vectors

- *Vectors $\neq$ Points*
- *Remember points can not be added*
- *If A and B are points A-B is a vector*
- *Vectors have form (x y z 1)*
- *Addition makes sense*

Translation in Homogenous Form

$$(x \ y \ z \ 1) \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ a & b & c & 1 \end{pmatrix} = (x+a \ y+b \ z+c \ 1)$$

- *Note that the homogenous component is preserved (* * * 1), and aside from the translation the matrix is I*

Putting it Together

$$\begin{pmatrix} R_1 & R_2 & R_3 & 0 \\ R_4 & R_5 & R_6 & 0 \\ R_7 & R_8 & R_9 & 0 \\ T_1 & T_2 & T_3 & 1 \end{pmatrix} = R.T$$

- *R is rotation and scale components*
- *T is translation component*

Order Matters

- *Composition order of transforms matters*
 - *Remember that basic vectors change so “direction” of translations changed*

$$(X \ Y \ Z \ 1) \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 2 & 3 & 4 & 1 \end{pmatrix} = (X \ Z \ -Y \ 1) \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 2 & 3 & 4 & 1 \end{pmatrix} = (X+2 \ Z+3 \ -Y+4 \ 1)$$

$$(X \ Y \ Z \ 1) \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 2 & 3 & 4 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} = (X+2 \ Y+4 \ Z+4 \ 1) \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & -1 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} = (X+2 \ Z+4 \ -Y-4 \ 1)$$

Matrix Summary

- *Rotation, Scale, Translation*
- *Composition of transforms*
- *The homogeneous form*